Cryptocurrency Volatility Prediction using Machine Learning

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Contents

1 Introduction **2**

2 Dataset Description **2**

3 Exploratory Data Analysis **2**
3.1 Distribution of Log Returns 3
3.2 Distribution of Rolling Volatility 3
3.3 Bitcoin Volatility Over Time 4
3.4 Bitcoin Price Trend 4
3.5 Feature Correlation 5

4 Feature Engineering **5**
4.1 Log Returns 5
4.2 Rolling Volatility 6
4.3 Target Variable 6

5 Model Development **6**

6 High-Level Design **6**

7 Low-Level Design **7**

8 Pipeline Architecture **7**

9 Conclusion **7**

1 Introduction

Cryptocurrency markets are highly volatile and experience rapid price fluctuations. Predicting volatility is important for traders, investors, and financial institutions because it helps in risk management and portfolio allocation.

This project aims to build a machine learning model that predicts future cryptocurrency volatility using historical market data such as prices, trading volume, and market capitalization.

Machine learning methods are well suited for this task because they can capture nonlinear relationships and complex patterns in financial data.

2 Dataset Description

The dataset contains historical market information for multiple cryptocurrencies. Each record represents daily trading data for a specific cryptocurrency.

The dataset includes the following variables:

- Open price
- High price
- Low price
- Close price
- Trading volume
- Market capitalization
- Cryptocurrency name
- Date

Additional engineered features were created to help the model learn patterns in market volatility.

3 Exploratory Data Analysis

Exploratory data analysis was performed to understand the statistical properties of cryptocurrency markets.

3.1 Distribution of Log Returns

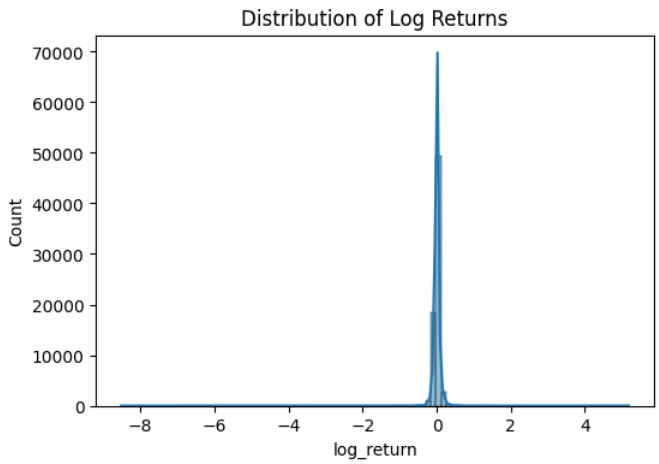

Figure 1: Distribution of Log Returns

The distribution shows that most returns are close to zero, while extreme price movements occur occasionally. This heavy-tailed behavior is typical for financial markets.

3.2 Distribution of Rolling Volatility

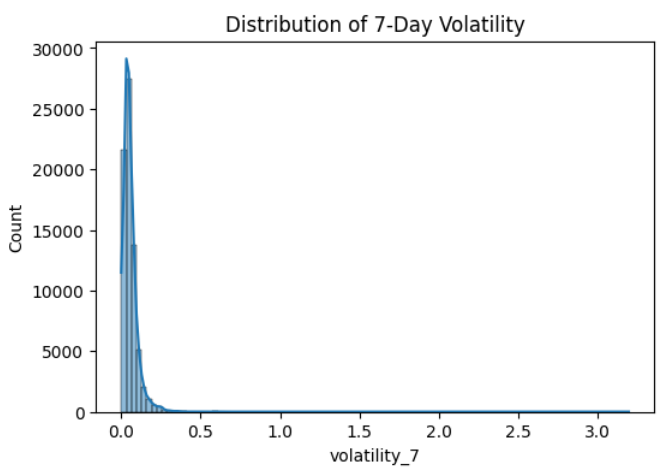

Figure 2: Distribution of Rolling Volatility

Volatility is highly skewed with occasional large spikes corresponding to major market events.

3.3 Bitcoin Volatility Over Time

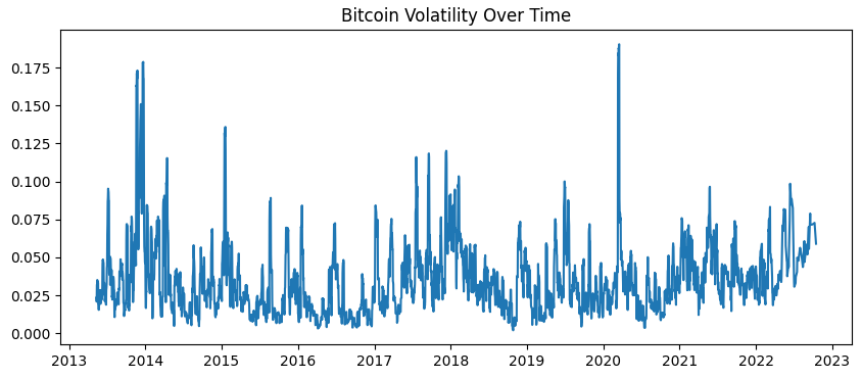

Figure 3: Bitcoin Volatility Over Time

The plot demonstrates volatility clustering, where periods of high volatility tend to persist.

3.4 Bitcoin Price Trend

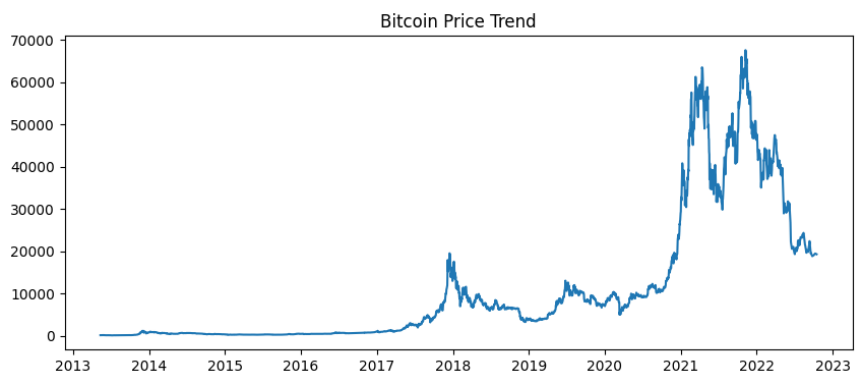

Figure 4: Bitcoin Price Trend

Bitcoin prices show multiple boom-and-bust cycles typical of cryptocurrency markets.

3.5 Feature Correlation

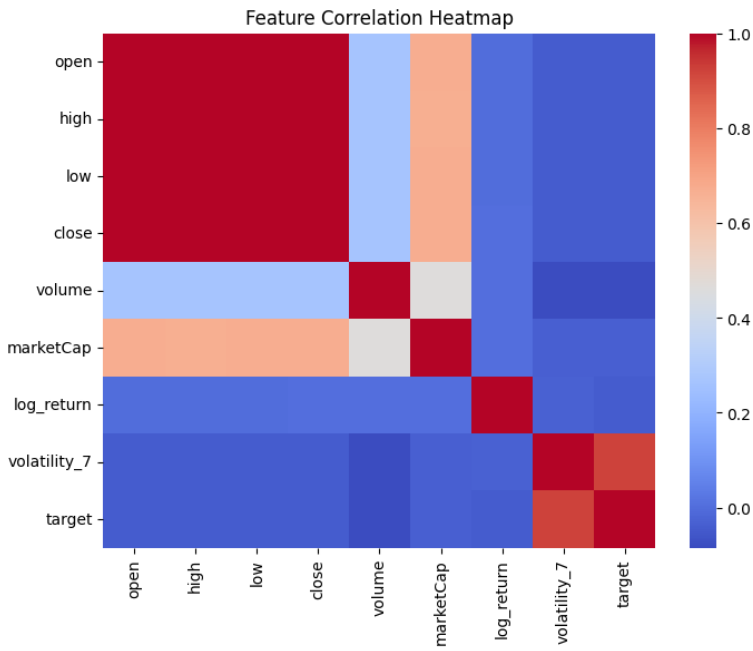

Figure 5: Feature Correlation Heatmap

Rolling volatility shows strong correlation with the prediction target, indicating that past volatility is useful for forecasting future volatility.

4 Feature Engineering

To capture important market behavior, several features were engineered.

4.1 Log Returns

Log returns measure relative price changes between consecutive days.

$$log_return = \log\left(\frac{close_t}{close_{t-1}}\right)$$

4.2 Rolling Volatility

Volatility was calculated as the rolling standard deviation of log returns over a 7-day window.

4.3 Target Variable

The model predicts future volatility by shifting the rolling volatility one step forward.

5 Model Development

An XGBoost regression model was used to predict cryptocurrency volatility.

XGBoost was selected because it handles nonlinear relationships effectively and performs well on structured financial data.

The dataset was split using a chronological time-based split to avoid data leakage.

Model performance was evaluated using the following metrics:

- R-squared score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

6 High-Level Design

The system consists of several major components.

- Data ingestion
- Data preprocessing
- Feature engineering
- Machine learning model training
- Prediction module
- Evaluation module

The system processes historical market data and generates volatility predictions using a trained machine learning model.

7 Low-Level Design

The implementation consists of the following steps:

1. Load dataset using pandas
2. Convert date column to datetime format
3. Sort data chronologically
4. Compute log returns
5. Calculate rolling volatility
6. Create prediction target
7. Perform time-based train-test split
8. Train XGBoost model
9. Evaluate model performance

8 Pipeline Architecture

The machine learning pipeline follows the following stages:

$$\text{Raw Market Data} \rightarrow \text{Data Cleaning} \rightarrow \text{Feature Engineering} \rightarrow \text{Train/Test Split} \rightarrow \text{Model Training} \rightarrow \text{Prediction} \rightarrow \text{Evaluation}$$

This pipeline ensures that historical information is used to predict future volatility without introducing data leakage.

9 Conclusion

This project developed a machine learning pipeline for predicting cryptocurrency volatility using historical market data.

Exploratory data analysis revealed important characteristics such as heavy-tailed return distributions and volatility clustering.

The XGBoost model leveraged these patterns to generate volatility predictions.

Future improvements could include incorporating additional technical indicators, sentiment analysis, or advanced time-series models.